

TCO — Handwritten Answer Sheet Evaluation Flowchart

End-to-End System Workflow Diagram with Standard Flowchart Elements & Multi-Agent Verification Loops

PRODUCT TCO — Teacher Co-Worker

SUBSYSTEM M Copy Checking Engine

ORCHESTRATOR Master Agent (Sole Teacher Interface)

FLOW ELEMENTS ISO 5807 Standard Symbols

DATA INTEGRITY: In-Memory RAM Processing (No Disk Retention)

GOVERNANCE Mandatory Teacher Final Approval Gate

LANE 1: INGESTION, IDENTITY & SCHEME LOCK

START: Evaluation Job Initiated

Teacher Submits Request via TCO Frontend

INPUT: Student Answer Sheet PDF

Original File: STU047.pdf + Question Paper + Max Marks
Teacher uploads student handwritten submission
Provides optional Question Paper & Exam Structure
Specifies Total / Maximum Marks (e.g., 50)

STU047.pdf

PROCESS: Parse Student Identity Key

Guarantees Student Processability from Source File

• Filename Parser: 'STU047.pdf' → student_number = 'STU047'
• Primary Key for all student lookups in TCO Student DB
• CRITICAL RULE: Temporary upload filenames are discarded

student_number = 'STU047'

SUBROUTINE: Master Agent Preconditions

Central Controller Verifies Prerequisite Issues

• Validates Answer Sheet PDF readability & DPI
• Verifies Question Paper & Maximum Marks
• Precondition Check: Marking Scheme Available?
• Teacher Communication Gateway: Sole user interface

Evaluate Inputs

D1: Teacher Provided Scheme?

Rubric Origin Check

NO: Generate

PROCESS: AI Generates Proposed Scheme

Question-by-Question Rubric Generation

• Drafts step points, partial marks & criteria
• Presents proposed rubric to Teacher via UI
• Awaits teacher verification & feedback

Review

D2: Scheme Approved?

Teacher Approval Gate

YES: Approved

PROCESS: Lock Immutable Scheme

One-Clicking Scheme Boundary

• Freezes question-wise point allocations & criteria
• Enforces strictness to assessment job ID
• HARD GATE: Copy checking CANNOT start prior to lock

Lock Rubric

DOCUMENT: Locked Marking Scheme

Immutable Policy Asset for Downstream Models

• Authorizes execution to proceed to Page Processing
• Evaluation model must strictly follow this rubric
• Precondition satisfied: Ready for In-Memory extraction

LANE 2: IN-MEMORY EXTRACTION & AUDIT (LOOP 1)

PROCESS: PDF In-Memory Processor

Ephemeral Page-by-Page Dispatcher

• Validates PDF structure, resolution, and page count
• Initiates sequential page raster (Page 1..N)
• ZERO permanent intermediate page-image storage

D3: Page N Available?

Page Iteration Loop

PROCESS: Render Page N into RAM

Strict In-Memory Execution Lifecycle

• Loads single page image directly into memory buffer
• No disk image retention; ephemeral lifecycle
• Dispatches in-memory raster to Vision AI model

Page in RAM

PROCESS: Extraction Model (Vision AI)

Multimodal Handwriting Perception (NOT Simple OCR)

• Understands handwritten layout, formulas, tables, and notes
• Contextual Question Number Detection (No hardcoded coordinates)
• Locates numbers in margin, above, circled, underlined
• Outputs: {page_no, q_no, student_answer, confidence, HOG}

Extracted Result + Original Page

PROCESS: Extraction Verifier (Model 2)

Independent Second AI Audit Engine

• Inputs: Extracted JSON + Original Page held in RAM
• Mission: 'Did extraction accurately read student writing?'
• Audits: Missing answers, transcription errors, hallucinations
• Checks question association & low-confidence flags
• RULE: Verifier does NOT silently edit; outputs feedback

Audit Check

D4: Extraction Correct?

Verification Loop 1 Gate

YES: Verified

PROCESS: Purge RAM & Store Page JSON

Ephemeral Image Clean-up

• IMMEDIATELY frees Page N image buffer from RAM
• Appends verified structured JSON to page buffer
• Advances loop counter: N = N + 1 → Return to D3

PROCESS: All-Page Aggregation

Multi-Page Synthesis & Normalization Engine

• Aggregates verified pages across entire answer sheet
• Stitches multi-page answers (e.g. Q3 on Page 1 & 2)
• Cross-page audits: missing questions, repeated attempts
• Normalizes question sequence and flags ambiguities

Aggregate Structure

DOCUMENT: Verified Student Submission

Trusted Ground-Truth Dataset (Strictly Un-Graded)

• Unified dataset: question numbers, answers, confidence
• Decouples perceptual transcription from academic evaluation
• Precondition met: Ready for Academic Evaluation

LANE 3: ACADEMIC GRADING & AUDIT (LOOP 2)

PROCESS: Ingest Evaluation Context

Assemble Required Academic Inputs

• 1. Question Paper / Exam Structure
• 2. Approved & Locked Marking Scheme
• 3. Maximum Marks per question & exam total
• 4. Verified Student Answers (from Lane 2)

PROCESS: Evaluation Agent (Model 3)

Domain-Specific Question Rubric Evaluation

• Evaluates answers strictly against locked marking scheme
• Step-Based Marking: Awards partial marks for working steps
• Calling Enforcement: Marks cannot exceed maximum marks
• Evaluates unanswered questions (0 marks + remarks)
• Outputs: Assigned Marks + Step Breakdown + Remarks
• RULE: Model follows locked scheme; CANNOT invent rules

Marks + Step Reasoning

PROCESS: Grading Verifier (Model 4)

Independent Quality Control & Audit Model

• Inputs: Grading Results + Question Paper + Locked Scheme
• Audit Mission: 'Are assigned marks fair and rubric-compliant?'
• Audits: Rubric compliance, calling check, reasoning validity
• Arithmetic Verification: Question step sums & overall total
• Flags unsupported deductions or unjustified additions
• RULE: Verifier does NOT alter marks; issues critique

Audit Evaluation

D5: Grading Correct?

Verification Loop 2 Gate

YES: Grading Validated

PROCESS: Synthesize Checked Copy

Visual & Data Overlay Assembly

• Combines student's original handwritten pages with AI marks
• Generates side-by-side view with step remarks
• Calculates verified total marks and percentages
• Prepares interactive copy for teacher review session

Overlay Data

DISPLAY: Complete Checked Copy

Interactive UI Presentation to Teacher

• Original Student Copy Pages + AI Overlays
• Q1: Student Answer (AI Marks: 1.0 / 1.0 [Correct])
• Q2: Student Answer (AI Marks: 0.5 / 2.0 [Partial step])
• Q3: Student Answer (AI Marks: 2.0 / 2.0 [Method correct])
• Total Marks: 41.5 / 50.0 (83%) + Remarks

Staged Payload

PROCESS: Stage for Master Agent

Broadcastation Handoff to Human Interface

• Master Agent receives checked copy payload
• Initiates teacher review session in TCO Frontend
• Prepares revision tracking buffer for edits

LANE 4: TEACHER REVIEW RE-MARK & PERSISTENCE

PROCESS: Teacher Review (HTL)

Human Inspection via TCO Frontend

• Teacher inspects complete checked copy in UI
• Reviews question-wise scores, remarks & page views
• TWO DECISION CHOICES:
• 1. [APPROVE] → Proceed to Final Gate
• 2. [REQUEST CHANGES] → Point or natural edits

D6: Teacher Approved?

Final Human Authority Gate

NO: Edit Request

SUBROUTINE: Re-marking

Selective Question Editing

• Teacher edits: 'Q3: 1.0, Q7: 3'
• Isolates ONLY Q2 and Q7
• Recalculates totals & percentage
• Updates copy for review again

Commit Approved Record

Commit Approved Record

DATABASE: TCO Student DB & Results

Committed ONLY After Final Teacher Approval

• Primary Identity: student_number ('STU047') matched to profile
• Assessment Log: 'EXAM-2026-MATH-011' (job_id: JOB_98412)
• question_wise_marks: {Q1: 1, Q2: 0.5, Q3: 1.0, ...}
• total_marks: 41.5 / 50.0 [percentage: 83.0%]
• teacher_approved: final_marks: 41.5 | status: 'TEACHER_APPROVED'
• feedback/remarks, audit logs & ISO-8601 timestamp stored
• SECURITY: Temporary upload filename is NEVER stored as key

Committed Record

OUTPUT: Final Evaluation Summary

Master Agent Dispatches Completion Report

• Confirms database persistence completed
• Student ID: STU047 (Verified against profile)
• Final Score: 41.5 / 50.0 (83.0%) [APPROVED]
• Delivers pedagogical feedback & remarks

Notify Teacher

END: Evaluation Workflow Completed

Results Archived & Teacher Notified

FLOWCHART SYMBOL LEGEND & ARCHITECTURAL FOUNDATIONS

START / END

Terminal

TERMINATOR SYMBOL
Represents the starting or ending boundary of the evaluation workflow.

DATA IN / OUT

Parallelogram

INPUT / OUTPUT SYMBOL
Represents external data ingestion (PDFs, pages) or UI display (checked copy).

PROCESS STEP

Rectangle

PROCESS SYMBOL
Represents an internal action, AI inference, validation or aggregation calculation.

DECISION SYMBOL

Diamond

DECISION SYMBOL
Represents a conditional branch (Yes/No) triggering verification retry loops.

SUBROUTINE

Double Border

SUBROUTINE SYMBOL
Represents a specialized sub-workflow (e.g., Selective Re-marking, Scheme Approval).

POLICY ASSET

Document

DOCUMENT ASSET
Represents an immutable or verified data asset (Locked Rubric, Verified Submission).

DATABASE

Cylinder

DATABASE PERSISTENCE
Represents persistent storage committed ONLY after explicit final teacher sign-off.

FEEDBACK

Dashed Arrow

FEEDBACK LOOP
Dashed arrows represent corrective re-evaluation & refinement cycles.

TCO — Flow Element Dictionary, Logic Matrices & Specifications

Comprehensive Reference of Flowchart Elements, Decision Truth Tables, JSON Contracts & Architectural Guarantees

1. STANDARD FLOWCHART ELEMENTS DICTIONARY & TCO SUBSYSTEM MAPPING

SYMBOL TYPE	ISO SHAPE	FLOW ELEMENT NAME	TCO ARCHITECTURAL FUNCTION	STAGE & ROLE IN EVALUATION WORKFLOW
TERMINATOR	Oval / Stadium Capsule	Start / End Boundary	Marks the external initiation (teacher submit) and termination (notification) of the copy checking job.	Start: Frontend submission End: Master Agent final summary delivery
INPUT / OUTPUT	Parallelogram (Skewed)	Data Input / UI Display	Represents external files ingested (PDF, Question Paper) and human-facing UI presentations (Checked Copy).	Ingestion of 'STU047.pdf' Presentation of AI question marks side-by-side with student writing
PROCESS	Rectangle	Action / Execution Step	Represents automated algorithmic tasks, AI vision transcription, aggregation logic, or arithmetic sums.	Identity parsing, In-Memory RAM page render, Vision AI extraction, All-Page aggregation, Final score totals
DECISION	Diamond (4-vertex Rhombus)	Conditional Evaluation Gate	Represents conditional branch points (Yes/No) that either allow flow forward or trigger corrective feedback loops.	D1: Scheme Provided? D2: Scheme Approved? D3: Page Loop D4: Extraction OK? D5: Grading OK? D6: Teacher Approved?
SUBROUTINE	Double-Bordered Rectangle	Predefined Sub-Workflow	Represents an autonomous multi-step sub-workflow that isolates and resolves complex specialized tasks.	Marking Scheme Approval Loop (interactive generation) Selective Re-marking Engine (targeted question recalculation)
DOCUMENT	Rectangle with Tab/Wave	Immutable Data Asset	Represents a verified, immutable state or policy document that binds downstream processing.	Locked Marking Scheme (enforces grading bounds) Verified Student Submission (un-graded ground truth)
DATABASE	Cylinder with Top Rim	Persistent Storage	Represents permanent database records that are committed strictly after final human sign-off.	TCO Student DB (primary student profile lookup) Assessment Results Store (marks, remarks, audit trail)
FEEDBACK LINE	Dashed Arrowed Path	Corrective Feedback Loop	Represents automated or human-in-the-loop critique returning to a prior stage for targeted re-evaluation.	Loop 1: Extraction audit retry Loop 2: Grading audit retry Loop 3: Teacher selective re-marking

2. DECISION LOGIC TRUTH TABLE & VERIFICATION RECOVERY MATRICES

GATE ID	DECISION QUESTION	EVALUATING AGENT	CRITERIA & VERIFICATION CHECKS	YES PATH (PASS)	NO PATH (FAIL / RETRY LOOP)
D1	Teacher Provided Scheme?	Master Agent	Checks teacher upload payload for pre-existing marking scheme document or rubric criteria.	Validate rubric & proceed directly to locking gate.	Trigger AI Scheme Generator to draft proposed rubric.
D2	Scheme Approved?	Teacher (via Frontend)	Teacher inspects proposed question-wise criteria, step allocations, and maximum marks in UI.	Lock Marking Scheme permanently; authorize In-Memory Page Processing.	Teacher provides correction feedback; AI regenerates improved scheme (Loop).
D3	Page N Available?	In-Memory Processor	Iterates sequentially through PDF pages: checks if current page index N <= total page count.	Render Page N image directly into RAM; trigger Extraction Model.	All pages processed; proceed to All-Page Aggregation & Synthesis.
D4	EXTRACTION CORRECT?	Extraction Verifier (Model 2)	Compares extracted JSON against original page in RAM: checks for missing answers, transcription errors, hallucinations, low confidence.	Page Verified: Purge Page N image from RAM; append JSON to submission buffer.	FAIL [Loop 1]: Generate structured error critique; re-run Extraction Model with context.
D5	GRADING CORRECT?	Grading Verifier (Model 4)	Audits grading results against locked scheme: checks rubric compliance, max mark limits, step credit validity, arithmetic sums.	Grading Passed: Synthesize Complete Checked Copy with AI mark overlays.	FAIL [Loop 2]: Issue marking discrepancy feedback; trigger Evaluation Agent re-grade.
D6	Teacher Approved?	Teacher (via Frontend)	Teacher reviews complete checked copy, inspected question marks (Q1: 1/1, Q2: 0.5/2), remarks, and overall total marks.	PASS: Authorize Database write; commit final approved marks to TCO Student DB.	FAIL [Loop 3]: Route teacher feedback to Selective Re-marking Engine (e.g. Q3=1, Q2=3).

3. INTER-STAGE DATA CONTRACTS & JSON OUTPUT SCHEMAS

EXTRACTION MODEL OUTPUT SCHEMA (Per Question)

Perceptual Ground Truth — Strictly UI-Graded

```
{  "page_number": 1,  "question_number": "Q3",  "question_position": {    "detected_layout": "margin_left_circled",    "visual_confidence": 0.98  },  "student_answer_raw": "E = mc^2 \n c = 3x10^8 m/s",  "answer_format": "mathematical_equation",  "source_region_bounding_box": [120, 450, 890, 680],  "extraction_confidence": 0.96,  "extraction_status": "VERIFIED_PASS",  "warnings": []}
```

EVALUATION AGENT OUTPUT SCHEMA (Per Question)

Rubric-Guided Scoring & Step Allocations

```
{  "question_number": "Q3",  "max_marks": 5.0,  "assigned_marks": 3.0,  "step_breakdown": [    {"step": "Formula stated correctly", "awarded": 1.0, "max": 1.0},    {"step": "Correct value substituted", "awarded": 1.0, "max": 1.0},    {"step": "Arithmetic calculation", "awarded": 1.0, "max": 2.0},    {"step": "Units and final answer", "awarded": 0.0, "max": 1.0}  ],  "evaluation_reasoning": "Formula and substitution are correct...",  "teacher_remarks": "Minor calculation slip in final step.",  "verification_audit_status": "AUDIT_PASSED"}
```

PERSISTED DATABASE RECORD SCHEMA

Permanent Storage — Committed Only After Final Approval

```
{  "student_number": "STU047",  // Parsed from STU047.pdf  "assessment_id": "EXAM_2026_MATH_01",  "job_id": "JOB_EVAL_98412",  "question_wise_marks": {    "Q1": 1.0,    "Q2": 0.5,    "Q3": 1.0,    ...  },  "total_marks_awarded": 41.5,  "maximum_marks_possible": 50.0,  "percentage": 83.0,  "teacher_approved_final_marks": 41.5,  "approval_status": "TEACHER_APPROVED",  "teacher_correction_log": [    {      "Q3_modified": 3.0 -> 1.0,      ...    }  ],  "timestamp": "2026-10-07T14:19:30Z"}
```

4. CORE ARCHITECTURAL PRINCIPLES, PRIVACY GUARANTEES & MODEL ABSTRACTION

MODEL PROVIDER ABSTRACTION

Minimal Model Quality is Reproducible; Pipeline is Stable

- Decoupled Architecture: Business workflow does not rely on a single model.
- Vision AI Interface: Standardized input/output abstraction.
- Current Prototype: Local Open Vision model.
- Production Swappable: Claude 3.5 Sonnet, GPT-4o, Gemini 1.5 Pro.
- Reasoning & Grading Interface: Standardized rubric reasoning schema.
- Swappable LLMs / fine-tuned rubric models for evaluation.
- Independent verification models for automated checks.
- Zero Vendor Lock-in: Models can be swapped without changing flow logic.

IN-MEMORY RAM PRIVACY POLICY

Strict ephemeral Page-by-Page Processing

- Strict In-Memory Lifecycle: Pages loaded strictly one-at-a-time into RAM.
- Lifecycle Sequence: Render Page N -> Vision AI Extract -> Audit -> Purge RAM.
- Zero Permanent Image Storage: Intermediate page rasters never touch disk.
- Student Data Protection: Eliminates unauthorized image caching and leaks.
- Resource Optimization: Memory footprint bounded to a single page image.
- High Throughput: In-memory pipeline processes multi-page sheets rapidly.
- Persistent Storage Bounded: Only final approved structured records stored.

MASTER AGENT RESPONSIBILITIES

Central Brain & Dual Human-Facing Interface

- Side Teacher Interface: Specialist AI models never message the teacher.
- Precognition Guard: Prohibits copy evaluation before marking scheme lock.
- Verification Loop Controller: Orchestrates Loop 1 (Extract) & Loop 2 (Grade).
- Synthesizer: Coordinates all-page aggregation into trusted ground truth.
- HTL Mediator: Receives teacher corrections & triggers selective re-marking.
- Persistence Guard: Enforces final approval gate before DB commit.
- Completion Dispatcher: Transmits certified score summary to teacher.

HUMAN-IN-THE-LOOP GOVERNANCE

Mandatory Human Authority Gate Before Persistence

- Complete Copy Transparency: Teacher inspects full student pages & AI marks.
- Direct Point Adjustments: Teacher can modify any question score.
- Natural Language Feedback: Supports requests like 'Q3 to 2 to be badle 1 do'.
- Selective Re-marking: Isolates affected questions; avoids full re-evaluation.
- Deterministic Recalculation: Question totals & percentages update instantly.
- Hard Persistence Lock: Database write prohibited without explicit sign-off.
- Audit Trail: Preserves full log of original AI marks vs teacher adjustments.